Chapter 12

Multifactor designs with repeated measurements.

1. The two-factor model with interaction.
2. Estimation.
3. Testing.
4. Sums of square decomposition.
5. Data example.
6. Random and mixed effects model.

The two-factor model with interaction

The following two-factor analysis is taken from Lawson, p. 64 f.

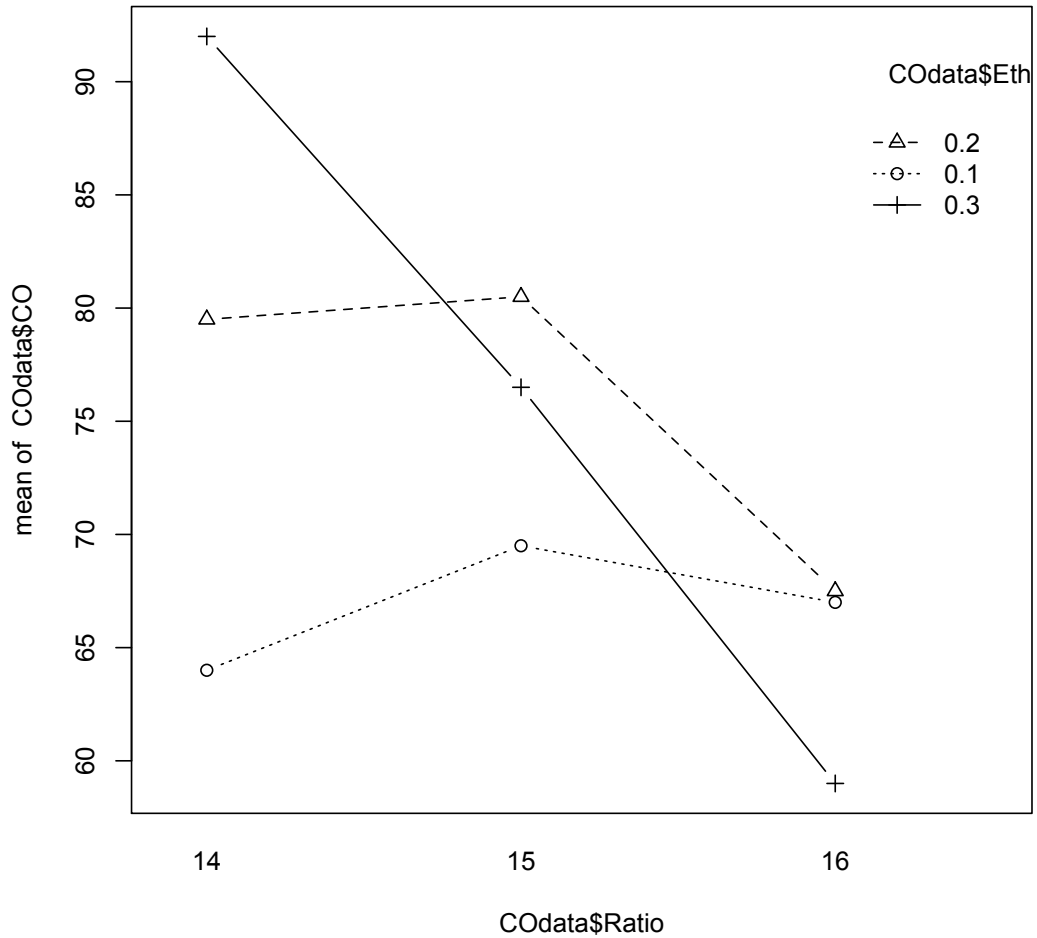
The experiment consists in burning an amount of fuel and determining the CO emissions released. Experimental unit is a portion of a standard fuel used for one run. The response Y is the CO emission concentration in grams/m^3. Factor A (three levels) is the amount of ethanol added to an experimental unit and Factor B is the fuel-to-air ratio (three levels) during the burn of fuel. Two replicate runs are done for each combination. Hence we have 18 observations in total.

The two-factor model with interaction

```
> library(daewr)
> COdata
   Eth Ratio CO
1  0.1    14 66
2  0.1    15 72
3  0.1    16 68
...
16 0.3    14 94
17 0.3    15 78
18 0.3    16 58
```

The two-factor model with interaction

The interaction plot indicates interaction between the factors.



The two-factor model with interaction

We formulate the linear models. We assume

$$Y_{ijk} \sim \mu_{ij} + \varepsilon_{ijk}, \quad 1 \leq i \leq r, ; 1 \leq j \leq s; 1 \leq k \leq K; n = rsK.$$

The ε_{ijk} are i.i.d. $N(0, \sigma^2)$. The new parameter K is the number of repetitions per factor combination. Remember that we assume a balanced design.

We consider the following linear hypotheses:

$$\begin{aligned} H_{AB} &: \mu_{ij} \quad \text{arbitrary for all } i, j; \\ H_{A+B} &: \mu_{ij} = \alpha_i + \beta_j; \\ H_A &: \mu_{ij} = \alpha_i; \\ H_B &: \mu_{ij} = \beta_j; \\ H_1 &: \mu_{ij} = \mu_0. \end{aligned}$$

Estimation

Estimation for under the models H_{A+B}, H_A, H_B and H_1 is analogue to what we have seen in Chapter 11. Averages need to be extended by summation over k.

So, for example for model H_{A+B} we have

$$\hat{\mu}_{ij} = \overline{Y}_{i\bullet\bullet} + \overline{Y}_{\bullet j\bullet} - \overline{Y}_{\bullet\bullet\bullet}$$

and

$$\tilde{\sigma}^2_{A+B} = \frac{1}{n-(r+s-1)} \sum_{i=1}^{s}\sum_{j=1}^{r}\sum_{k=1}^{K}(Y_{ijk} - \overline{Y}_{i\bullet\bullet} - \overline{Y}_{\bullet j\bullet} + \overline{Y}_{\bullet\bullet\bullet})^2.$$

We note that $\tilde{\sigma}^2_{A+B} =: MS(A+B)$ is the unbiased estimator of the variance under model H_{A+B}.

Estimation

By considering the factor combinations of factor A and B as the levels of a new single factor with $k = rs$ levels, we can apply the theory of Chapter 10 for H_{AB} and obtain for this case:

$$\hat{\mu}_{ij} = \overline{Y}_{ij\bullet}$$

and

$$\tilde{\sigma}^2_{AB} = \frac{1}{n - rs} \sum_{i=1}^{s} \sum_{j=1}^{r} \sum_{k=1}^{K} (Y_{ijk} - \overline{Y}_{ij\bullet})^2.$$

Here $\tilde{\sigma}^2_{AB} = MSR$ is the unbiased estimator of the variance under model H_{AB}.

Estimation

We remark that under H_{AB} the μ_{ij} can be written as $\mu_{ij} = \mu_0 + \alpha_i^0 + \beta_j^0 + \gamma_{ij}^{00}$, where

$$\sum_{i=1}^{r} \alpha_i^0 = 0; \quad \sum_{j=1}^{s} \beta_j^0 = 0; \quad \sum_{i=1}^{r} \gamma_{ij}^{00} = \sum_{j=1}^{s} \gamma_{ij}^{00} = 0.$$

The estimators for the effects are then given as

$$\begin{aligned}
\hat{\mu}_0 &= \overline{Y}_{\bullet\bullet\bullet}; \\
\hat{\alpha}_i^0 &= \overline{Y}_{i\bullet\bullet} - \overline{Y}_{\bullet\bullet\bullet}; \\
\hat{\beta}_j^0 &= \overline{Y}_{\bullet j\bullet} - \overline{Y}_{\bullet\bullet\bullet}; \\
\hat{\gamma}_{ij}^{00} &= \overline{Y}_{ij\bullet} - \overline{Y}_{i\bullet\bullet} - \overline{Y}_{\bullet j\bullet} + \overline{Y}_{\bullet\bullet\bullet}.
\end{aligned}$$

It is clear they satisfy the constraints and add up to $\overline{Y}_{ij\bullet}$. The γ_{ij} are the interaction effects. When γ_{ij} are equal to zero, we have an additive model.

Sums of square decomposition

Denote p_A, p_B, p_{AB}, ... the projections on the linear spaces L_A, L_B, L_{AB}, ... We also use the spaces $L_{A^*} = L_A \ominus L_1$ and $L_{B^*} = L_B \ominus L_1$ from the previous chapter. We recall that $p_{A+B} = p_{A^*} + p_{B^*} + p_1$. Then we have

$$p_{AB} = (p_{AB} - p_{A+B}) + p_{A^*} + p_{B^*} + p_1.$$

and thus

$$I - p_{AB} = (I - p_1) - p_{B^*} - p_{A^*} - (p_{AB} - p_{A+B})$$

and thus

$$(I - p_1) = p_{B^*} + p_{A^*} + (p_{AB} - p_{A+B}) + (I - p_{AB}).$$

Sums of square decomposition

The spaces corresponding to the r.h.s. of

$$(I - p_1) = p_{B^*} + p_{A^*} + (p_{AB} - p_{A+B}) + (I - p_{AB}).$$

to L_{B^*}, L_{A^*}, $L_{AB} \ominus L_{A+B}$ and $L_{AB}^{\perp}$ and are orthogonal. Hence by Pythagoras we have

$$\begin{aligned}
&\|Y - p_1(Y)\|^2 \\
&= \|p_A(Y) - p_1(Y)\|^2 + \|p_B(Y) - p_1(Y)\|^2 \\
&\quad + \|p_{AB}(Y) - p_{A+B}(Y)\|^2 + \|Y - p_{AB}(Y)\|^2
\end{aligned}$$

or

$$SST = SSA + SSB + SS(AB) + SSR.$$

In terms of degrees of freedom:

$$\underbrace{rsK - 1}_{n-1} = (r-1) + (s-1) + \underbrace{(rs - (r+s-1))}_{(r-1)(s-1)} + \underbrace{rsK - rs}_{n-rs}.$$

Sums of square decomposition

Scaling by degrees of freedom gives MSA, MSB, MS(AB) and MSR which we will need below for the F-tests.

We also obtain

$$R^2 = \frac{SSA + SSB + SS(AB)}{SST}.$$

Tests

We consider 3 types of tests.

We first test for absence of interactions.

This is $H_{A+B}(= H_{0AB}) : \gamma_{ij} = 0$ under H_{AB}. The corresponding F-statistic is

$$F_{0AB} = \frac{MS(AB)}{MSR}.$$

Under the null the statistic is distributed as $F_{(r-1)(s-1),n-rs}$.

We reject if $F_{0AB} > F_{(r-1)(s-1),n-rs;1-\alpha}$.

Tests

When there are no interactions, we may test for factors A and B separately. This is $H_{0A}\colon \alpha_i = 0$ against $H_{aA}\colon \alpha_i \neq 0$ for some i under the model H_{AB}.

The corresponding F-statistic is

$$F_{0A} = \frac{MSA}{MSR}.$$

Under the null the statistic is distributed as $F_{r-1,n-rs}$.

We reject if $F_{0A} > F_{r-1,n-rs;1-\alpha}$.

The test F_{0B} is analogue.

Data example continued

In our CO data the ANOVA confirms the presence of an interaction effect.

```
> mod=aov(CO~Eth*Ratio,data=COdata)
> summary(mod)
            Df Sum Sq Mean Sq F value   Pr(>F)
Eth          2  324.0   162.0   31.36 8.79e-05 ***
Ratio        2  652.0   326.0   63.10 5.07e-06 ***
Eth:Ratio    4  678.0   169.5   32.81 2.24e-05 ***
Residuals    9   46.5     5.2
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Data example continued

A look at `model.tables` shows also why an interaction is reasonable: minimizing marginally over factors, doesn't lead to a global minimum.

```
> model.tables(mod,type="means")
Grand mean
72.83333
Eth
  0.1   0.2   0.3
66.83 75.83 75.83
Ratio
  14   15   16
78.5 75.5 64.5
 Eth:Ratio
     Ratio
Eth   14   15   16
  0.1 64.0 69.5 67.0
  0.2 79.5 80.5 67.5
  0.3 92.0 76.5 59.0
```

Data example continued

Alternatively we can look at the effects:

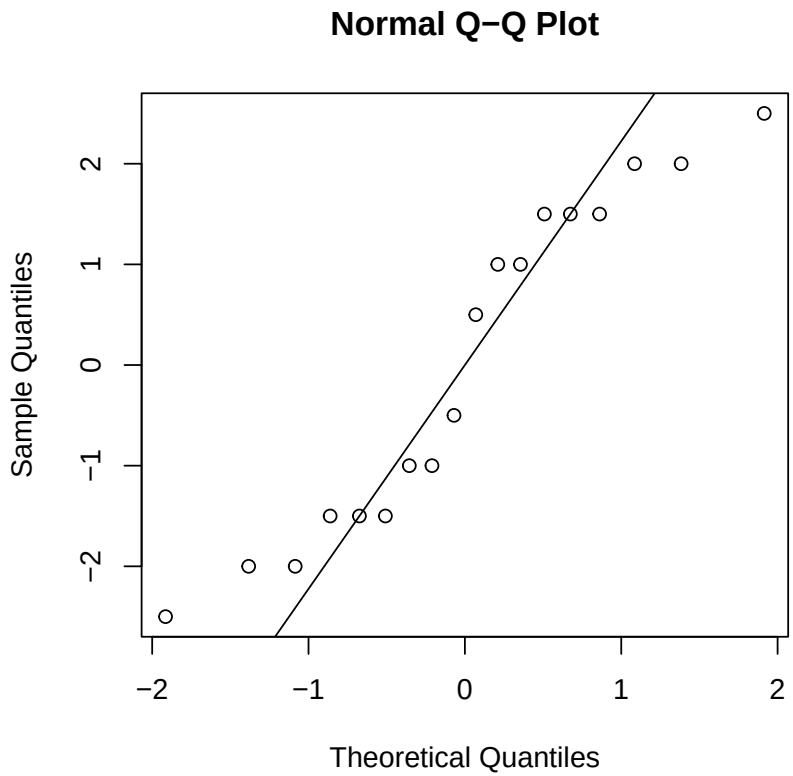
```
> model.tables(mod,type = "effects")
Tables of effects

 Eth
Eth
0.1 0.2 0.3
 -6   3   3

 Ratio
Ratio
    14     15     16
 5.667  2.667 -8.333

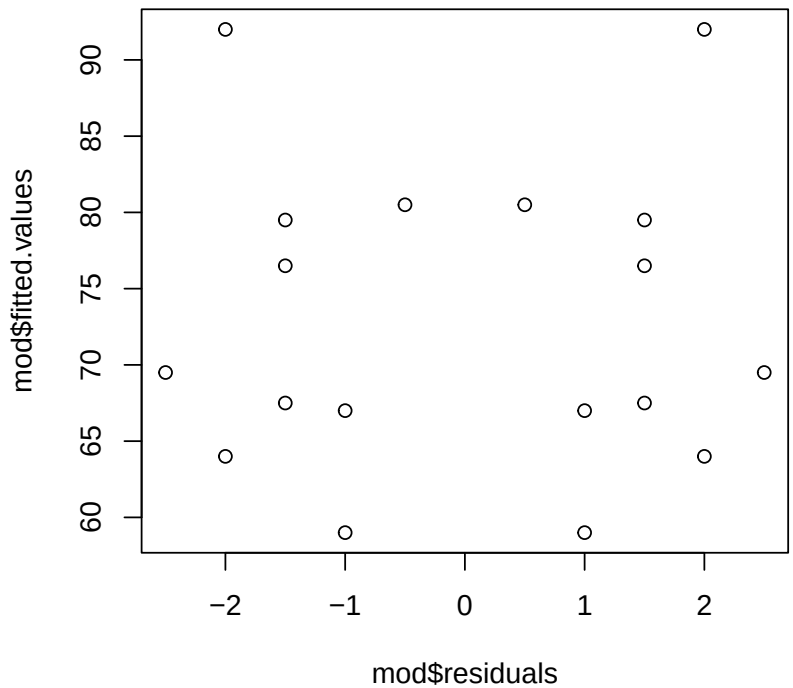
 Eth:Ratio
     Ratio
Eth    14   15   16
  0.1 -8.5  0.0  8.5
  0.2 -2.0  2.0  0.0
  0.3 10.5 -2.0 -8.5
```

Data example continued



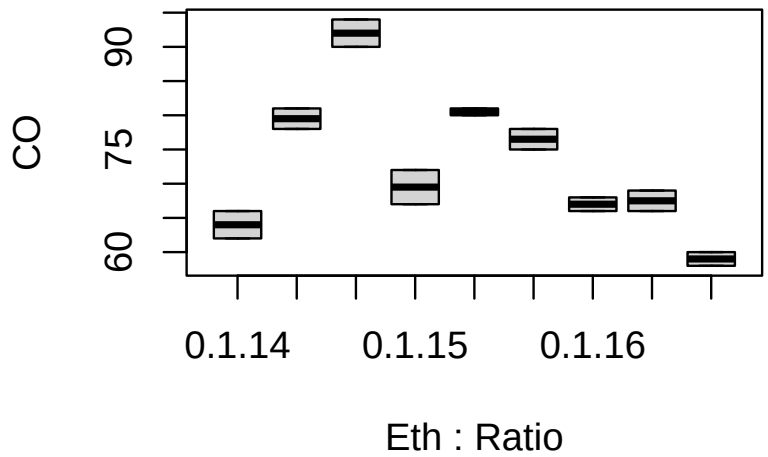
The qq-plot might indicate a violation of normality, but the Shapiro-Wilk normality test gives a p-value of 0.074 and hence doesn't reject the null (normality) at 5% significance level.

Data example continued



The plot of residuals vs fitted values doesn't appear suspicous. (The symmetry is due to the 2 observations per factor combination).

Data example continued



To check the homogenous variance assumption we proceed in a heuristic manner and make a box plot with the only $K = 2$ repetitions per factor combination. The hight of the box correpsonds to the spread between the observations. Non of them seems to particularly stick out.

Mixed and random effects

In some problems the levels of either factor may have been chosen from a variety of possible levels, so that the effect due to such a factor may be considered random.

Example:
Suppose we are interested in the natural degree of variation in sodium content across brands of beer.

There are dozens of brands of beer and by time and budget constraints we only can test a small number p.

We select p brands randomly and test K bottles of beer for each brand.

Mixed and random effects

The example above has only one factor. If we have two (or more) factors and if both (all) factors contribute random effects, we call this a random effects model. If only one (some) factor contributes random effects, we call it a mixed effect model.

The mixed effect model is defined via

$$Y_{ijk} = \mu + \alpha_i + B_j + G_{ij} + \varepsilon_{ijk},$$

with i, j and k varying over the usual index set. Here B_j, G_{ij} and ε_{ijk} are independent normally distributed variables with respective variances σ_B^2, σ_G^2 and σ^2.

Mixed and random effects

The hypothesis of no effect of factor B is formalized as $H_{0B}: \sigma_B^2 = 0$. The hypothesis for the effect of factor A to be non-significant, H_{0A}, remains the usual one.

It is customary to test H_{0A} and H_{0B} only if the no-interaction hypothesis $H_{0G}: \sigma_G^2 = 0$ cannot be rejected.

Like for the fixed effects model, the no-interaction hypothesis $H_{0AB}: \sigma_G^2 = 0$ is based on the F-Ratio

$$F_{AB} = \frac{MS(AB)}{MSR}.$$

We reject the null if $F_{AB} > F_{(r-1)(s-1);n-rs;1-\alpha}$.

Mixed and random effects

We notice that the expected mean squares differ from the fixed effects model. One can show the following (exercise):

Fixed effects:

$$EMSA = \sigma^2 + \frac{sK}{r-1}\sum_{i=1}^{r}\alpha_i^2;$$

$$EMSB = \sigma^2 + \frac{rK}{s-1}\sum_{j=1}^{s}\beta_j^2;$$

$$EMS(AB) = \sigma^2 + \frac{K}{(s-1)(r-1)}\sum_{i=1}^{r}\sum_{j=1}^{s}\gamma_{ij}^2;$$

$$EMSR = \sigma^2.$$

Mixed and random effects

On the other hand we have (exercise):

Mixed effects:

$$
\begin{aligned}
EMSA &= \sigma^2 + K\sigma_G^2 + \frac{sK}{r-1}\sum_{i=1}^{r}\alpha_i^2; \\
EMSB &= \sigma^2 + K\sigma_G^2 + r\sigma_B^2; \\
EMS(AB) &= \sigma^2 + K\sigma_G^2; \\
EMSR &= \sigma^2.
\end{aligned}
$$

Mixed and random effects

This motivates why $F_A = \frac{MSA}{MS(AB)}$ and $F_B = \frac{MSB}{MS(AB)}$ are used as corresponding tests for H_{0A} and H_{0B}.

Under the null we show below that

$$F_A \sim F_{r-1,(r-1)(s-1)};$$
$$F_B \sim F_{s-1,(r-1)(s-1)}.$$

Mixed and random effects

To justify this approach, we plug in the definition of Y_{ijk} into the corresponding sums of squares. It can be easily seen that

$$MS(AB)$$
$$= \sum_{i=1}^{r}\sum_{j=1}^{s}\sum_{k=1}^{K}(G_{ij} + \overline{\varepsilon}_{ij\bullet} - (\overline{G}_{i\bullet} + \overline{\varepsilon}_{i\bullet\bullet}) - (\overline{G}_{\bullet j} + \overline{\varepsilon}_{\bullet j\bullet}) + (\overline{G}_{\bullet\bullet} + \overline{\varepsilon}_{\bullet\bullet\bullet}))^2.$$

Note that compared to fixed effects we have to replace the variables $\overline{\varepsilon}_{ij\bullet} \overset{\text{iid}}{\sim} N(0, \frac{\sigma^2}{K})$ with $G_{ij} + \overline{\varepsilon}_{ij\bullet} \overset{\text{iid}}{\sim} N(0, \frac{\sigma^2}{K} + \sigma_G^2)$. Since these variables are the building blocks for the above sum of squares, and since for fixed effects we have $MS(AB) \sim \sigma^2 \chi^2_{(r-s)(s-1)}$, we conclude that in the current setup we have

$$MS(AB) \sim (\sigma^2 + K\sigma_G^2)\chi^2_{(r-s)(s-1)}.$$

Mixed and random effects

For MSR nothing changes:

$$MSR = \sum_{i=1}^{r}\sum_{j=1}^{s}\sum_{k=1}^{K}(\varepsilon_{ijk} - \overline{\varepsilon}_{ij\bullet})^2 \sim \sigma^2\chi^2_{n-rs}.$$

Moreover, MSR and $MS(AB)$ remain independent. This is because we can write again the sums of squares as norms of projections. Then independence of numerator and denominator comes on the one hand, since we do projections in orthognal spaces (as with fixed effects) and on the other hand since the G's in the numerator are independent of the ε's in the denominator.

Under the $H_{0G}: \sigma^2_G = 0$ we obtain that the ratio is scale-free and hence it has the desired distribution.

Mixed and random effects

For MSA we get

$$MSA = \sum_{i=1}^{r}\sum_{j=1}^{s}\sum_{k=1}^{K}(\alpha_i + \overline{G}_{i\bullet} - \overline{G}_{\bullet\bullet} + \overline{\varepsilon}_{i\bullet\bullet} - \overline{\varepsilon}_{\bullet\bullet\bullet})^2$$
$$= K\sum_{i=1}^{r}\sum_{j=1}^{s}(\alpha_i + \overline{G}_{i\bullet} - \overline{G}_{\bullet\bullet} + \overline{\varepsilon}_{i\bullet\bullet} - \overline{\varepsilon}_{\bullet\bullet\bullet})^2.$$

By similar arguments as before, we see that under H_{0A}

$$MSA \sim (\sigma^2 + K\sigma_G^2)\chi^2_{r-1}$$

and that MSA and $MS(AB)$ are independent. For the corresponding F-statistic it is important that the variances cancel, in order to make them distribution free. For MSB the same type of argument works.

Data example

In this example we consider a random effects model. Hence we have two factors, which are both effects are considered random.

The data is contained in the `nmle` package and is calles `Pastes`. The data are coming from *deliveries of a chemical paste product contained in casks where, in addition to sampling and testing errors, there are variations in quality between deliveries . . . As a routine, three casks selected at random from each delivery were sampled and the samples were kept for reference. . . . Ten of the delivery batches were sampled at random and two analytical tests carried out on each of the 30 samples.*

Data example

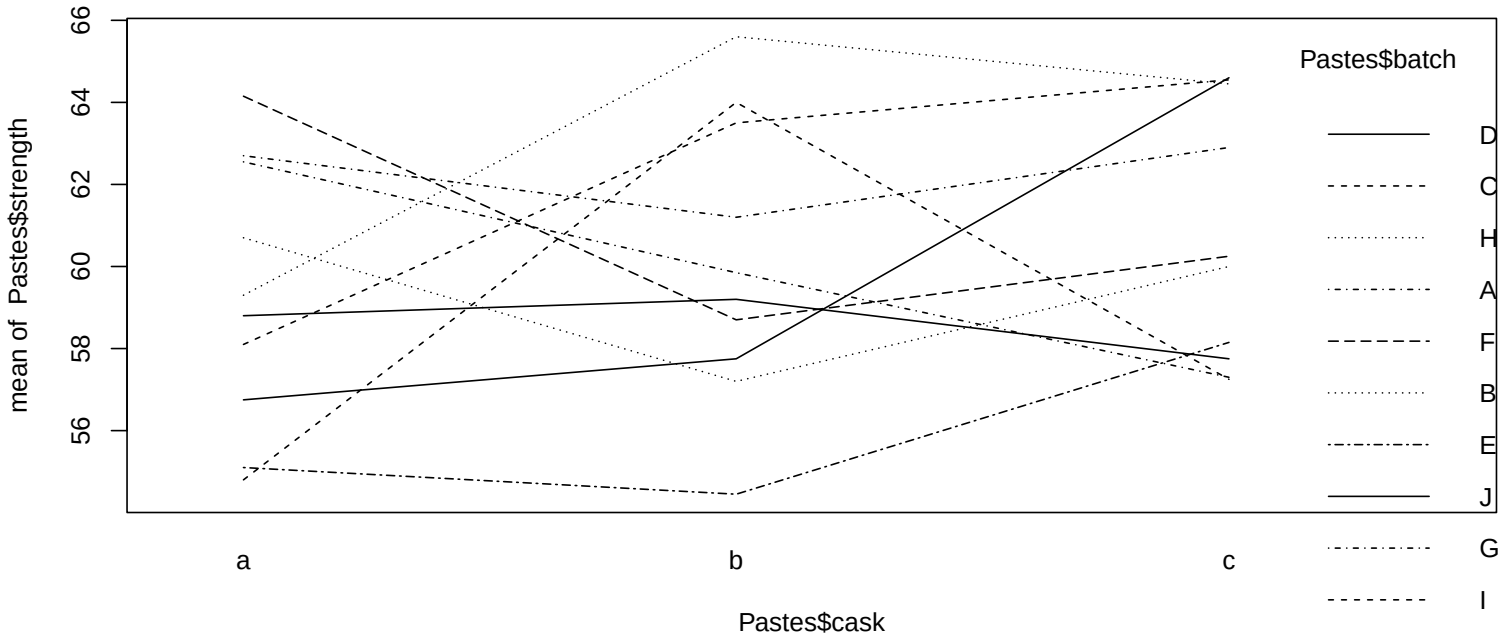
	strength	batch	cask	sample
1	62.8	A	a	A:a
2	62.6	A	a	A:a
3	60.1	A	b	A:b
4	62.3	A	b	A:b
...				
...				
58	59.2	J	b	J:b
59	58.9	J	c	J:c
60	56.6	J	c	J:c

Data example

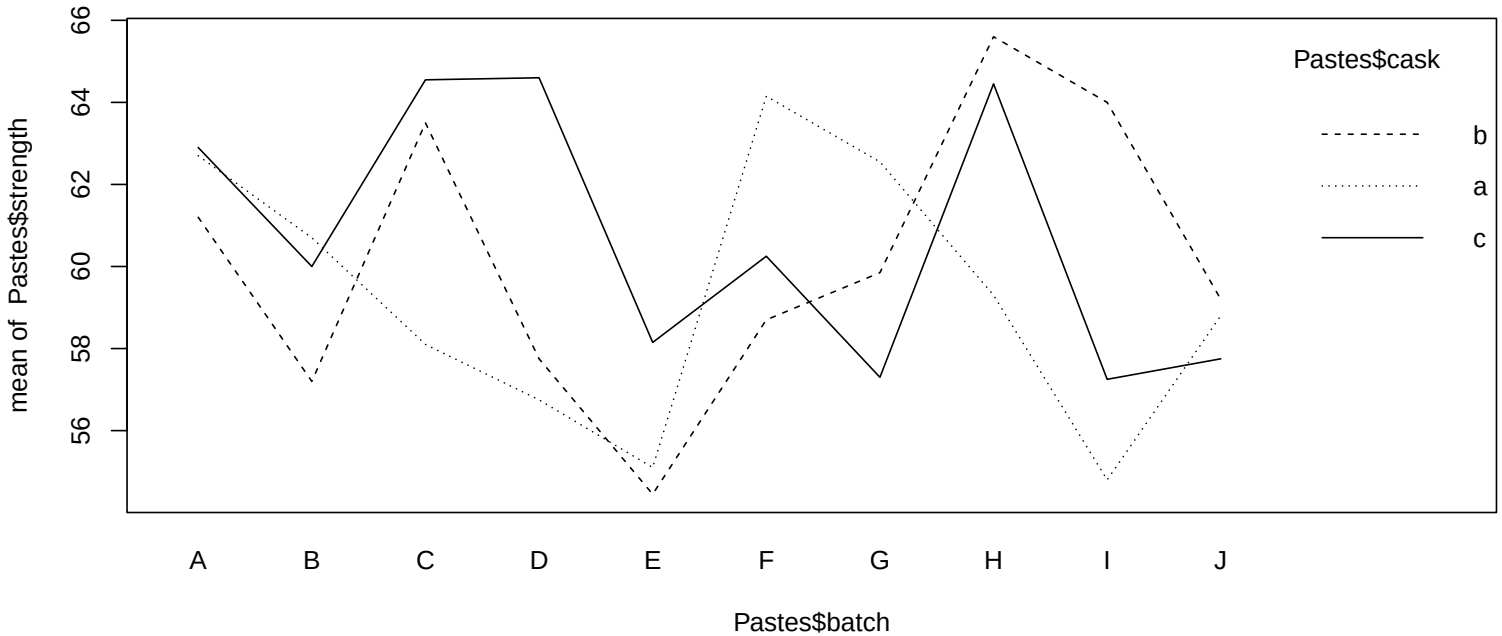
Both variables, `batch` and `cask` are randomly chosen from a big number of possible choices. We hence consider a two-way layout with random effects. We have a balanced design with $K = 2$ and $r = 10$ batches and $s = 3$ casks.

We start with interaction plots. By visual inspection an additive model seems not appropriate ...

Data example



Data example



Data example

When testing for the lack of interaction, we can do the usual ANOVA and test for no-interaction as if we had a fixed design:

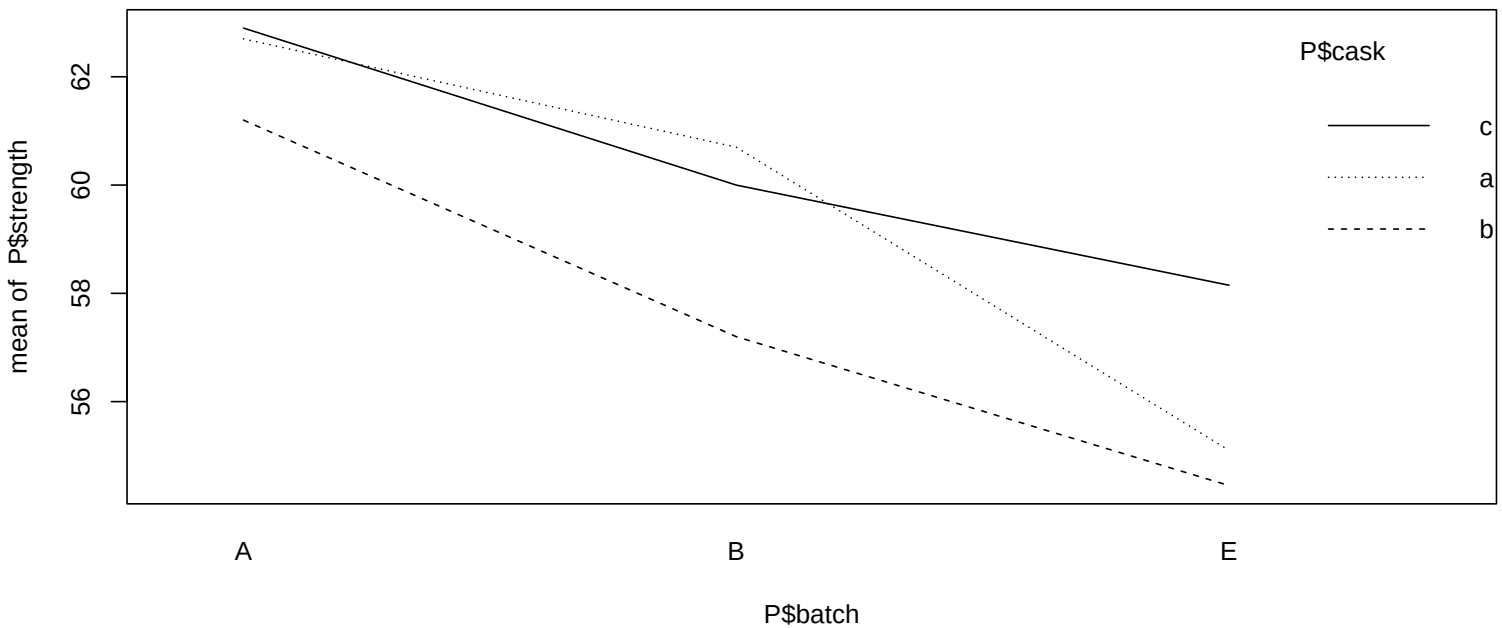
```
mod=aov(strength~batch*cask,data=Pastes)
summary(mod)
             Df Sum Sq Mean Sq F value   Pr(>F)
batch         9  247.4  27.489   40.55 2.28e-14 ***
cask          2   20.6  10.279   15.16 2.82e-05 ***
batch:cask   18  330.3  18.353   27.07 8.87e-14 ***
Residuals    30   20.3   0.678
```

According to this we have strong support for $H_{0G}: \sigma_G = 0$ to be violated.

Data example

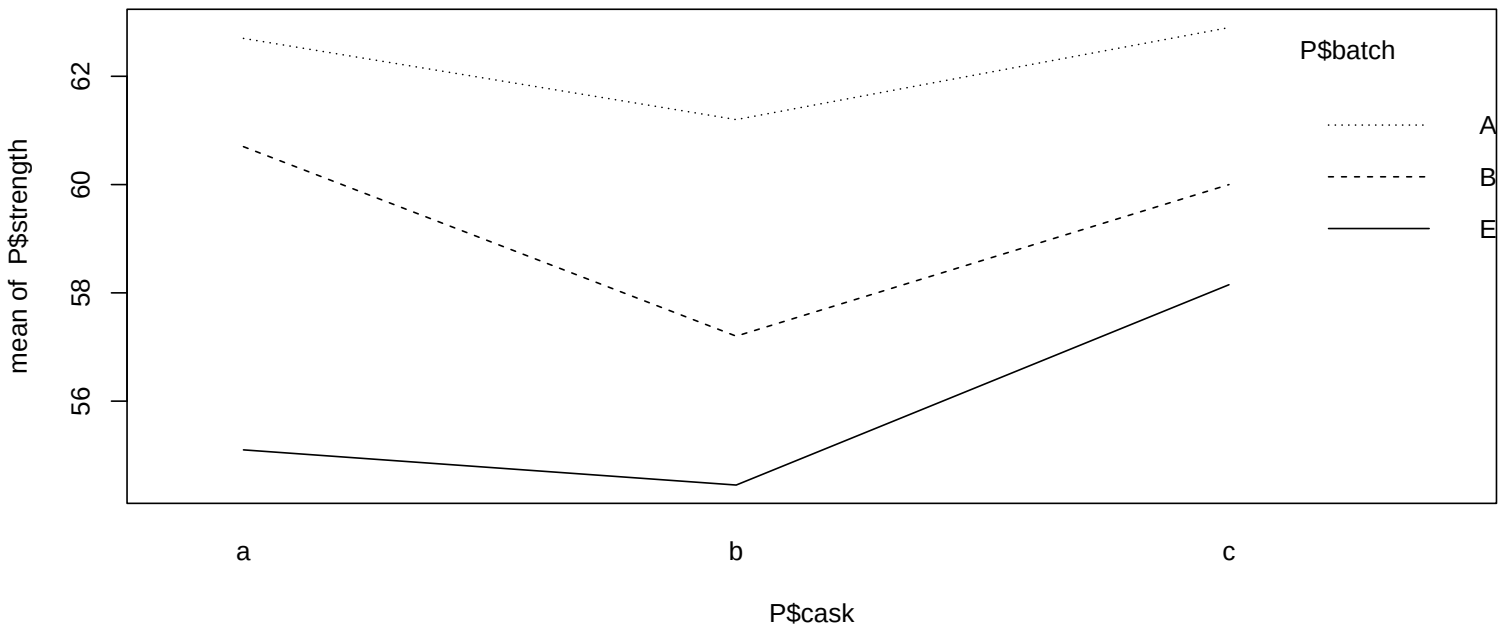
For illustrative purposes, suppose we had selected only batches A, B and E ($n = 18$).

Here the interaction plots look like this:



Data example

and this:



Data example

Now we have not that strong evidence for an interaction:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
batch	2	121.79	60.90	77.52	2.12e-06	***
cask	2	23.48	11.74	14.95	0.00138	**
batch:cask	4	9.30	2.33	2.96	0.08128	.
Residuals	9	7.07	0.79			

Let me stress that I was deliberately selecting batches! So generally this is of course not a valid approach.

However, let's ignore this and consider this as an example where $\sigma_G^2 = 0$ is reasonable.

Now it makes sense to test $H_{0A}: \sigma_A^2 = 0$ or $H_{0B}: \sigma_B^2 = 0$. Would we consider a fixed effects model we would reject either null.

Data example

In random effects model we consider

$$\frac{MSA}{MS(AB)} = \frac{60.9}{2.33} \quad \text{and} \quad \frac{MSB}{MS(AB)} = \frac{11.74}{2.33}.$$

Under the respective null, these ratios are both distributed as $F_{2,4}$. The corresponding p-values are $p = 0.0051$ and $p = 0.0807$.

According to this analysis, we don't get enough support for H_{0B} being violated, if we test at a 5% significance level.